Australian Government
Department of Industry, Science and Resources

Guiding principles for a global quantum ecosystem informed by science

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Introduction

The principles guide how the Multilateral Dialogue on Quantum (MDQ) works, collaborates and supports the development of a global quantum information science and technology (QIST) ecosystem.

We developed them in collaboration with the 12 other member countries of the MDQ. () Shared values underpin the principles, including:

- freedom of inquiry
- openness
- transparency
- honesty
- equity
- reciprocity
- fair competition
- objectivity
- accountability
- protection and enforcement of intellectual property
- democratic ideals.

The principles are an officials-level statement.

Guiding principles

1. **Promote scientific collaboration and the exchange of ideas** to support quicker scientific discovery in QIST.

Many scientific questions remain for QIST to reach its full potential. International cooperation among scientists is key to jointly addressing the global challenges at the frontier of QIST, accelerating discovery,

maintaining leadership for our ecosystems.

2. **Incorporate subject-matter expertise** in policy discussions and decisions.

The general unfamiliarity with QIST among the public and policymakers lends itself to hype and misconceptions. The group can leverage QIST subject-matter expertise for policy discussions to facilitate decisions based on the best available science.

3. **Share best practices and coordinate outreach** in joint effort to grow the QIST talent base.

The demand for QIST talent is currently outpacing supply. Sharing best practices for increasing awareness of and training in QIST will enable more people to have the opportunity to participate. Coordinated outreach efforts are mutually beneficial and expand the reach of any one nation.

4. **Promote research security, align and reinforce technology protection measures, and support a fair marketplace,** to create a vibrant and trusted global QIST industry.

The QIST industry is globally interconnected. This ecosystem can be enhanced with a trusted supply chain and developing international standards that enable the adoption of new quantum technologies. It can be safeguarded by promoting research security and through multilateral protection measures informed by an understanding of the technology and its progress.

5. **Plan for the deployment of quantum-resistant cyber infrastructure, such as quantum-resistant cryptography,** to responsibly address the risks of QIST.

Any system that uses existing public-key cryptography standards could be vulnerable to an attack by a future large-scale quantum computer. To mitigate this risk, the group should alert decision makers to plan for transitioning cyber infrastructure to quantum-resistant cryptography.

6. **Increase quantum awareness and readiness** in the development and use of quantum-based technologies and applications.

Awareness and understanding of the quantum domain and the inherent benefits and vulnerabilities of quantum-based technologies and applications concerns all aspects of society.

7. **Encourage the discovery of use cases of QIST** for the benefit of society.

The most significant applications of QIST are likely still to be discovered through fundamental research and development. The discovery of applications and use cases of QIST can be further strengthened by connecting researchers with practitioners, pioneering industries, and end-users through cooperation and partnerships. Together, the group can contribute to creating an environment that encourages and enables the discovery of use cases that are consistent with shared values and that benefit society.

Statement by

Members of the Multilateral Dialogue on Quantum:

- Australia
- Canada
- Denmark
- Finland
- France
- Germany
- Japan
- Korea
- Netherlands
- Sweden
- Switzerland
- United Kingdom
- United States.

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Email

Critical.Technology@industry.gov.au (https://web.archive.org/web/20260313111457/mailto:Critical.Technology@industry.gov.au)

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Department of Industry, Science and Resources

General Enquiries

+61 2 6213 6000

Industry House,
10 Binara Street,
Canberra

Department of Industry
ABN: 74 599 608 295

ACKNOWLEDGEMENT OF COUNTRY

Our department recognises the First Peoples of this Nation and their ongoing cultural and spiritual connections to the lands, waters, seas, skies, and communities.

We Acknowledge First Nations Peoples as the Traditional Custodians and Lore Keepers of the oldest living culture and pay respects to their Elders past and present. We extend that respect to all First Nations Peoples.

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